MOHAMED ABOELGHAR

Professor & Senior Researcher | Remote Sensing & GIS | Agricultural Applications

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Phone: +201000405188 | Location: Cairo, Egypt

ORCID | Google Scholar | LinkedIn | Portfolio

PROFESSIONAL SUMMARY

Professor and Senior Researcher with 30+ years of expertise in Remote Sensing, GIS, and AI-driven geospatial modelling for agricultural applications. Principal Investigator on 15+ national and international funded projects covering crop monitoring, yield prediction, water productivity assessment, applications of remote sensing and GIS for smart farming and land cover mapping across Egypt, Sudan, Iraq, and Japan. Author of 50+ peer-reviewed publications. Trilingual: Arabic (native), English (fluent), Japanese (conversational).

CORE COMPETENCIES

- Remote Sensing & Earth Observation: Satellite Data Analysis (Sentinel-2, Landsat, MODIS, SPOT, SAR)
- GIS & Geospatial Platforms: ArcGIS Pro | ENVI | QGIS | Google Earth Engine (GEE)
- AI & Machine Learning: Random Forest | Deep Learning | Object-Based Image Analysis (OBIA)
- Agricultural RS Applications: Crop Monitoring | Yield Prediction | NDVI/LAI Analysis | Precision Agriculture
- Water & Land Management: Water Productivity Assessment | LULC Mapping | Desertification Monitoring
- Research Leadership: Principal Investigator | Project Management | Multidisciplinary Team Leadership
- Programming & Tools: IoT Integration | Big Geospatial Data Management

PROFESSIONAL EXPERIENCE

Professor - Head of Agricultural Applications, Soil & Marine Sciences Division | 07/2025 - Present

National Authority for Remote Sensing and Space Sciences (NARSS), Cairo, Egypt

- Lead and supervise national research programs in Remote Sensing and GIS for agriculture
- Serve as Principal Investigator for local and international funded projects
- Develop short- and medium-term research strategies and ensure quality assurance of scientific outputs

Professor - Head of Agricultural Applications Department | 09/2019 - 06/2025

National Authority for Remote Sensing and Space Sciences (NARSS), Cairo, Egypt

- Planned and managed departmental research activities; led multidisciplinary research teams
- Delivered postgraduate, graduate, and professional training courses in RS and GIS
- Coordinated project execution, reporting, and stakeholder engagement for 8+ funded projects

Assistant Professor - Supervisor, Agricultural Applications Department | 09/2013 - 08/2019

National Authority for Remote Sensing and Space Sciences (NARSS), Cairo, Egypt

- Co-Principal Investigator on 6 agricultural remote sensing projects
- Taught undergraduate and postgraduate courses; contributed to data analysis and publications

Researcher | 06/2008 - 08/2013

National Authority for Remote Sensing and Space Sciences (NARSS), Cairo, Egypt

- Conducted applied research in agricultural Remote Sensing and GIS
- Delivered professional training programs, supported satellite data processing and technical reporting

Lecturer | 04/2005 - 05/2008

Center for Environmental Remote Sensing (CEReS), Chiba University, Japan

- Developed global geodatabases for land cover products (GLCNMO) for JAXA
- Contributed to MODIS-based global land cover mapping; published in International Journal of Digital Earth

Research Assistant | 10/1995 - 03/2001

National Authority for Remote Sensing and Space Sciences (NARSS), Cairo, Egypt

- Supported agricultural remote sensing research projects
- Assisted with satellite data collection, pre-processing, and analysis

Additional Academic and Consulting Roles

- Adjunct Professor – Ain Shams University, Faculty of Art, Egypt (2019-2025): Introductive and Advanced Remote Sensing & GIS
- Adjunct Professor - Zewail City for Science and Technology, Egypt (2016-2018): Remote Sensing & GIS for ecosystem monitoring
- Adjunct Professor - Menofia University, Faculty of Agriculture, Egypt (2014-2015): Remote Sensing in Plant Pathology
- Consultant - Arab Organization for Agricultural Development, League of Arab States (2015): Technical advisory & training in Sudan and Arab countries

SELECTED RESEARCH PROJECTS

- **2026-2027 |** Development of an Integrated National Framework for Monitoring Vegetation Health in Egypt through Time-Series Remote Sensing, Spectral Vegetation Indices, and Artificial Intelligence (Phase I: Al-Fayoum governorate)
- **2025-2026 |** Crop Pattern Monitoring of Dakahlia and Damietta Governorates Using High-Resolution Satellite Imagery and AI -- NARSS, Egypt
- **2024-2025 |** Multi-Source Remote Sensing Data and AI for Crop Type Mapping in Northern Nile Delta and Middle Egypt -- NARSS, Egypt
- **2023-2024 |** Winter Agricultural Crops Survey -- NARSS & Ministry of Agriculture, Egypt
- **2022-2023 |** Integrated Geospatial Database for Sustainable Agricultural Development in Fayoum Governorate -- NARSS & Fayoum Governorate, Egypt
- **2020-2022 |** Management and Applications of Remote Sensing Big Data in Old and Reclaimed Lands for Sustainable Agricultural Development -- NARSS, Egypt
- **2019-2020 |** Integrated Geospatial System for Periodic Monitoring of Egyptian Export Crops Using RS and GIS -- NARSS, Egypt
- **2018-2019 |** Integrated Crop Monitoring and Estimation of Water Productivity in Egyptian Nile Delta -- NARSS, Egypt
- **2016-2017 |** Remote Sensing and GIS for Crop and Water Management in Nile Valley and Delta -- NARSS, Egypt
- **2015-2016 |** Assessment of Desertification in Waset and Misan Provinces, Iraq Using RS and GIS -- Arab Organization of Agricultural Development
- **2014-2015 |** Water Harvesting in Kassab Area, Garadef, Sudan -- Arab Organization of Agricultural Development
- **2012-2014 |** Monitoring and Estimation of the Area of Main Strategic Crops in Egypt -- NARSS, Egypt
- **2009-2011 |** Modeling Yield Prediction for Main Cereal Crops in Egypt Using Multi-Source RS Data -- NARSS, Egypt
- **2005-2008 |** Global Land Cover Mapping (GLCNMO) -- Japanese Space Agency (JAXA), Japan

EDUCATION

- **Ph.D. in Remote Sensing | Chiba University, Chiba, Japan | March 2005**

Dissertation: Agricultural Land Monitoring and Crop Yield Prediction Using Remote Sensing

- **M.Sc. in Plant Pathology | Menoufia University, Egypt | October 1998**

Dissertation: Studies on Some Citrus Diseases Using Remote Sensing Techniques

- **B.Sc. in Plant Pathology | Menoufia University, Egypt | May 1995**

SELECTED PUBLICATIONS

- **2026:** Multi-sensors Remote Sensing and Machine Learning Techniques Applications in Agriculture. Egypt. J. Bot., 66(1), 1-17
- **2025:** Magnesium Oxide Nanoparticles Enhance Growth Performance, Spectral Vegetation Indices, Yield Quantity and Quality of Green Onion. Russ J Plant Physiol 72, 124
- **2025:** Delineation of Site-Specific Management Zones to Enhance Nutrient Status, Growth, and Quality of Green Onion in Newly Reclaimed Area, Ismailia, Egypt. Front. Sustain. Food Syst. 9:1508115
- **2022:** Crop Yield Prediction Using Multi Sensors Remote Sensing (Review Article). The Egyptian Journal of Remote Sensing and Space Science, 25(3), 711-716
- **2021:** Integrated Method for Rice Cultivation Monitoring Using Sentinel-2 Data and Leaf Area Index. The Egyptian Journal of Remote Sensing and Space Science, 24(3), 431-441
- **2014:** Spectral Wheat Yield Prediction Modeling Using SPOT Satellite Imagery and Leaf Area Index. Arabian Journal of Geosciences, 7:465-474
- **2013:** Rice Yield Forecasting Models Using Satellite Imagery in Egypt. The Egyptian Journal of Remote Sensing and Space Sciences, 16, 125-131
- **2011:** Production of Global Land Cover Data - GLCNMO. International Journal of Digital Earth, 4(1), 22-49
- **2004:** Agricultural Land Monitoring in the Egyptian Nile Delta Using Landsat Data. International Journal of Environmental Studies, 61(6), 651-657

Full publication list available on Google Scholar Profile

PROFESSIONAL DIPLOMAS

- Diploma in Project Management - NARSS (2011)
- Diploma in Strategy and National Security - Nasser Military Academy (2023)
- Diploma in Crisis Management & Decision Making - Nasser Military Academy (2023)
- Diploma in Leadership Development - Helwan University (2024)

LANGUAGES

- Arabic: Native / Bilingual
- English: Fluent
- Japanese: Conversational